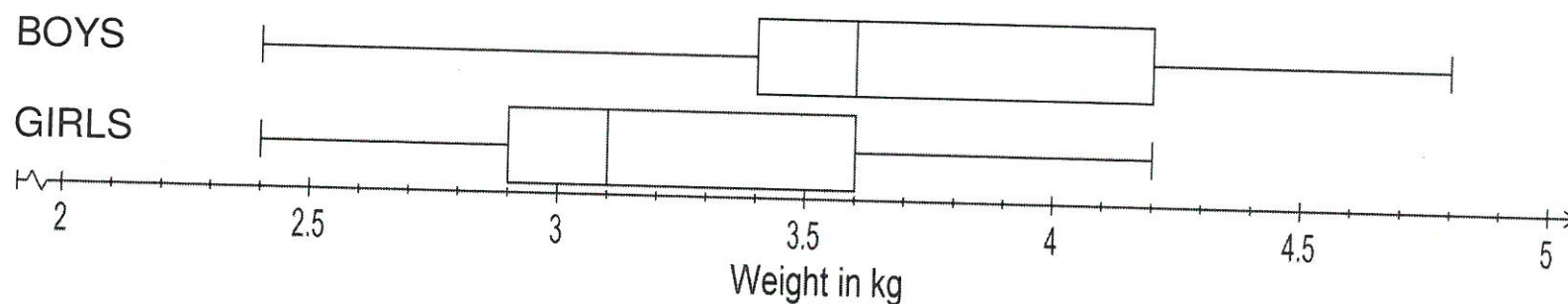


Comparing Parallel Boxplot

Consider the following parallel box plots showing the weights of newborn boys and girls.



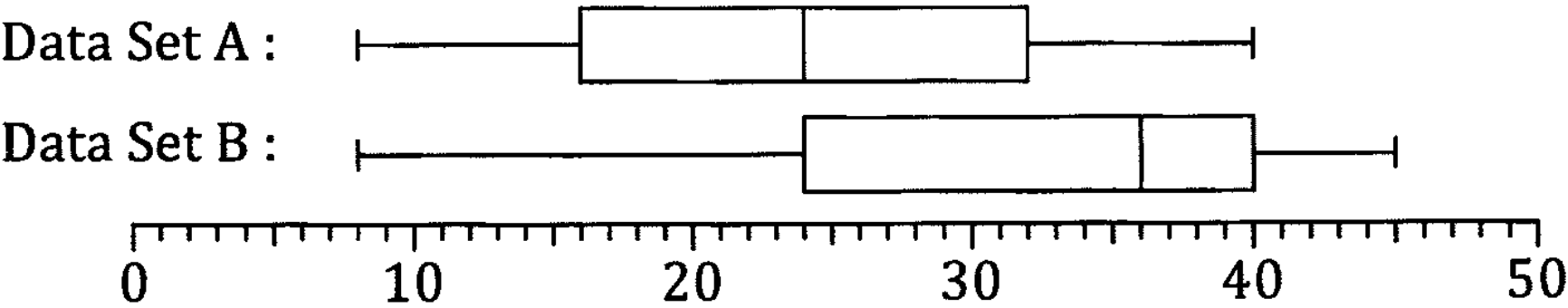
The question that often arises is “are newborn boys heavier than newborn girls?”

To answer this question let us compare the two data sets displayed by the box plots given above.

- STEP 1:** Compare the location of the data sets by comparing the respective medians.
- STEP 2:** Compare the dispersion or spread of the data sets by comparing the range, that is the overall spread and the interquartile range, that is the box lengths of the box plots.
- STEP 3:** Compare the shape of the data sets, that is symmetry or asymmetry and comment.
- STEP 4:** Look and check for outliers.
- STEP 5:** General conclusions about the data sets should be summarised briefly.

- Comparison of Location:** The median birth weight of the boys is 3.6 kg which is greater than the median weight of the girls 3.1 kg.
- Comparison of Spread:** The interquartile ranges(boys 0.8 kg, girls 0.7 kg) are reasonably similar as shown by the length of the boxes. The overall range of birth weights is greater for the boys as shown by the distances between the ends of the two whiskers for each box plot. Boys' range is 2.4 kg, girls 1.8 kg.
- Comparison of Shape:** Both distributions of birth weights appear to be skewed to the right the weights of the boy newborns is slightly more skewed to the right than for the girls.
- Check for outliers:** Neither data set shows any data points of extreme value, hence no outliers.
- General Conclusion:** Overall the two data sets are generally distributed in a similar way with the distribution of the boys' birth weights located to the right of the girls' birth weights. The median birth weight of the girls is less than the lower quartile of the boys' birth weights. Hence we can confidently conclude that based on the given data sets newborn boys are heavier than newborn girls.

Compare the distributions shown in the box plots below.



The median for data set B, 36, is much higher than that of data set A, 24.

Both data sets have a lowest score of 8 but set B has the greater highest score, 45, compared to 40 for set A.

Data set B has a range of 37 compared to 32 for set A.
Both data sets have an interquartile range of 16.

The box plot for set A is symmetrical and each quarter of the scores span 8 marks. On the other hand the longer left whisker of set B and the greater part of the box being to the left of the median suggest the set B marks are skewed to the left.

The median score in set A is the same as the lower quartile score in set B.

The top 25% of the marks from set B exceeded the top mark in set A.



Compare location.



Compare spread.



Compare shape.



Anything else of
relevance.